Lab 4: free vibration of a 1-DOF system

From a sensor-cart record to mass, stiffness, natural frequency, and damping

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Free response | natural frequency | damping from decay

Why this experiment matters

Vehicles

A suspension rings after a bump. Engineers choose mass, stiffness, and damping to control settling.

bump → ring-down

Machines

A motor, pump, or tool can amplify vibration near a natural frequency and shorten component life.

disturbance → free response

Structures

Floors, towers, aircraft panels, and instruments must settle quickly without excessive acceleration.

disturbance → free response

Lab 4 asks the same engineering question: after a displacement and release, how fast does the system oscillate and how quickly does it settle?

The laboratory question and the evidence

We will change one physical parameter at a time.

- Mass m: cart, hex bars, and dampers
- Stiffness k: two selected springs
- Damping c: 0, 2, or 4 eddy-current dampers

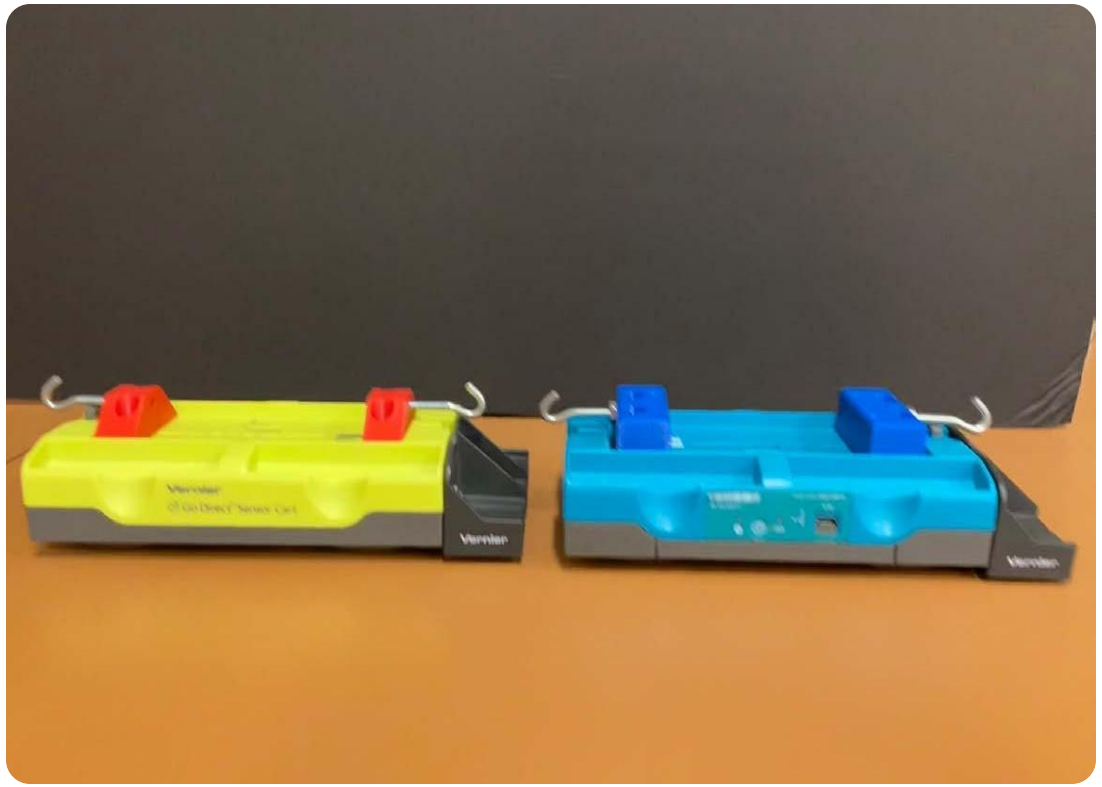
Measured record

$x(t)$ → f_n , ζ

Position versus time becomes evidence for natural frequency and damping ratio.

Prediction before collection: increasing m lowers frequency; increasing k raises frequency; increasing damping shortens the visible ring-down.

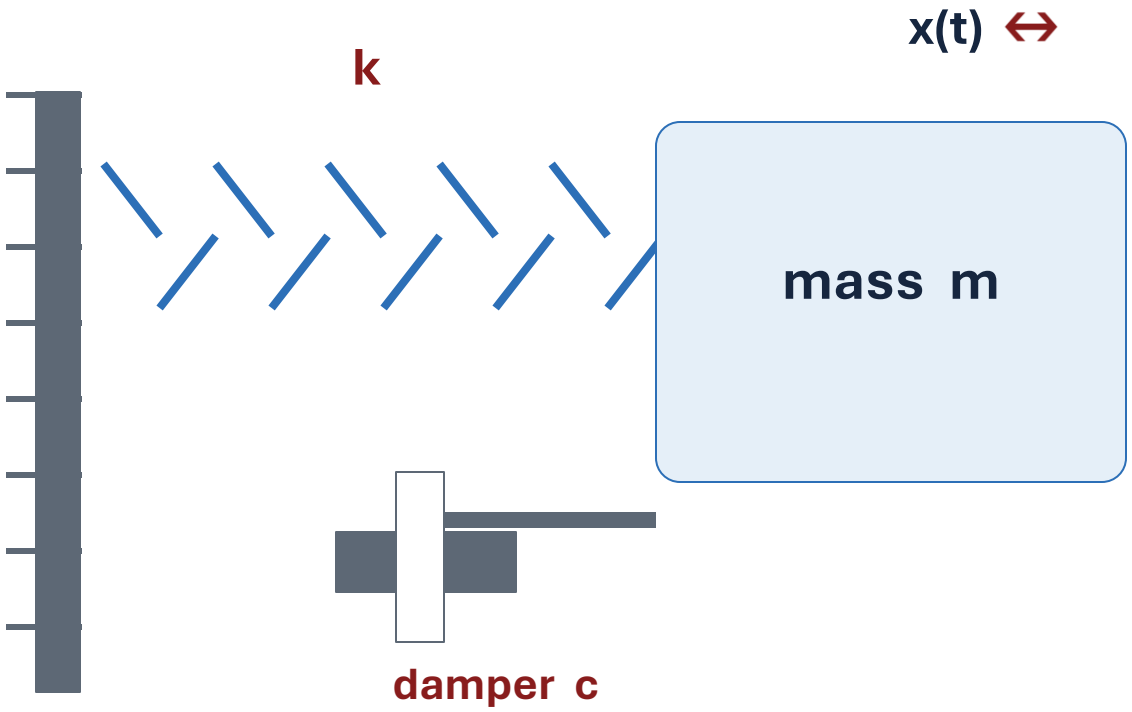
The apparatus records position, not “damping”



- The wheel encoder converts wheel rotation to cart position x(t).
- Graphical Analysis supplies time stamps and exports a CSV file.
- Springs create restoring force; magnets create velocity-dependent resistance.
- Your analysis converts the measured record into physical parameters.

Important: record mass, spring pair, damper count, initial displacement, sample rate, and file name for every run.

The hardware becomes a spring–mass–damper model



$$m\,\ddot{x} + c\,\dot{x} + k\,x = 0$$

- x(t): cart displacement from equilibrium [m]
- m: total moving mass [kg]
- c: viscous-equivalent damping coefficient [N·s/m]
- k: equivalent stiffness of both springs [N/m]

A free-body diagram gives the equation of motion



$$\Sigma F_x = m\,\ddot{x} \;\Rightarrow\; -kx - c\dot{x} = m\,\ddot{x}$$

Both spring and damping forces oppose the displacement or velocity that created them. The zero on the right side of the standard form means no external force acts after release.

Each parameter changes a different feature of the record

symbol	role	force/response	visible effect
m	**inertia**	resists acceleration	larger m → slower oscillation
k	**restoring stiffness**	pulls toward equilibrium	larger k → faster oscillation
c	**energy dissipation**	removes mechanical energy	larger c → faster decay

Without damping, the cart exchanges energy forever

$$m\ddot{x} + kx = 0$$

$$\omega_n = \sqrt{(k/m)} \quad [\text{rad/s}]$$

$$x(t) = A\cos(\omega_n t - \phi)$$

$$f_n = \omega_n/(2\pi) \quad [\text{Hz}]$$

ω_n is the undamped natural angular frequency. f_n counts cycles per second. A and ϕ come from the initial displacement and velocity.

Dimensional check: k/m has units s^{-2}, so $\sqrt{(k/m)}$ has units rad/s.

Natural frequency is a property of the system

Double the mass

f_n becomes $1/\sqrt{2} = 0.707$ times the original

Double the stiffness

f_n becomes $\sqrt{2} = 1.414$ times the original

Double m and k

f_n stays unchanged

$$f_n \propto \sqrt{k} \quad \text{and} \quad f_n \propto 1/\sqrt{m}$$

Energy explains the repeated motion

$$E(t) = \tfrac{1}{2}m\dot{x}^2 + \tfrac{1}{2}kx^2$$

turning point

x = ±A

spring energy maximum
speed = 0

equilibrium

x = 0

spring energy minimum
speed maximum

With c = 0, total mechanical energy stays constant. Real damping removes energy on every cycle.

Damping removes energy fastest when the cart moves fastest

$$F_d = -c\,\dot{x}$$

$$P_d = F_d\,\dot{x} = -c\,\dot{x}^2$$

- The damping force reverses when velocity reverses.
- The damper does no work at a turning point because $\dot{x} = 0$.
- The magnitude of energy loss is largest near equilibrium, where speed is largest.
- The magnetic dampers approximate viscous damping over the operating range.

We model several loss mechanisms with one equivalent coefficient c. The model is useful only if the measured envelope resembles an exponential.

Damping ratio classifies the response

$$\ddot{x} + 2\zeta\omega_n \dot{x} + \omega_n^2 x = 0$$

$$\zeta = c / (2\sqrt{km})$$

$0 < \zeta < 1$

underdamped

oscillates and decays

$\zeta = 1$

critical

fastest nonoscillatory return

$\zeta > 1$

overdamped

slow nonoscillatory return

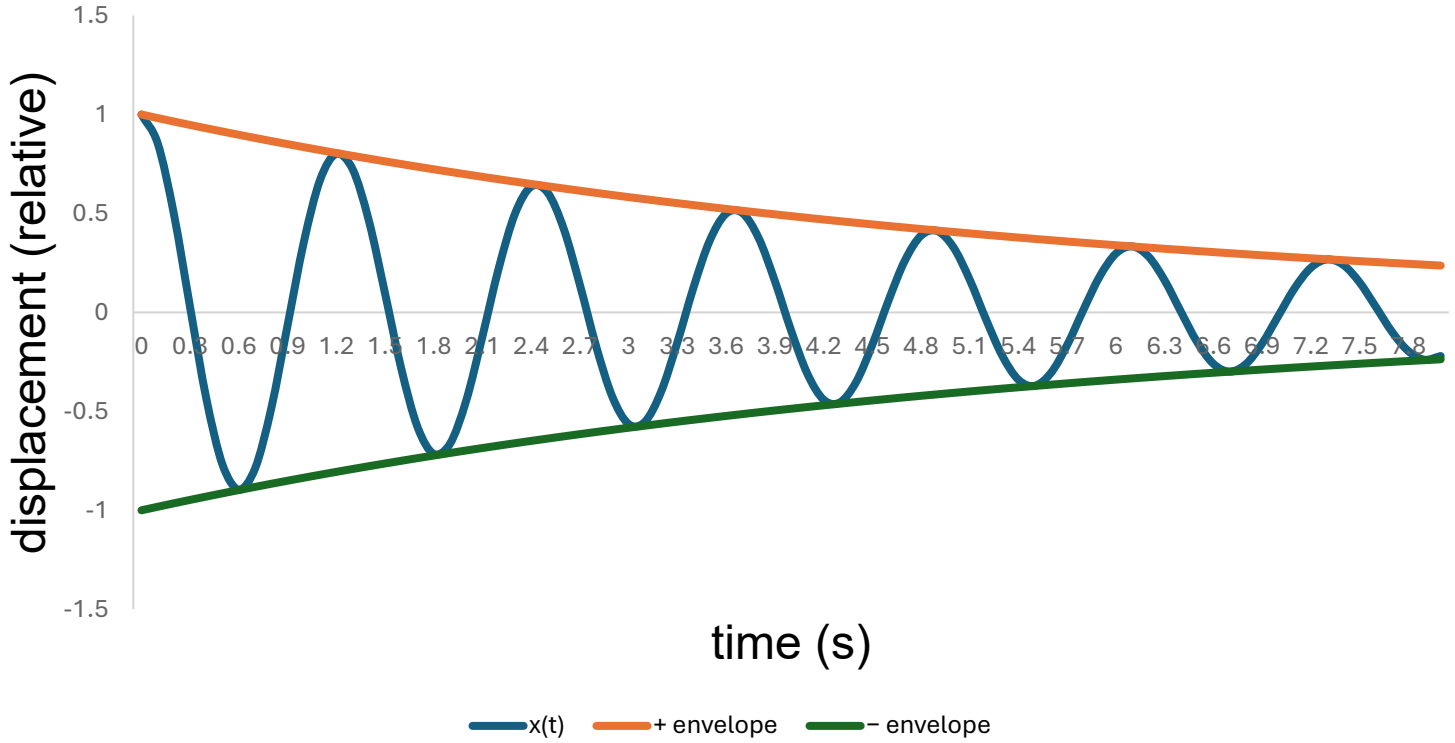
Lab 4 operates in the underdamped regime

$$x(t) = A \exp(-\zeta \omega_n t) \cos(\omega_d t - \phi)$$

$$\omega_d = \omega_n \sqrt{(1 - \zeta^2)}$$

- A: response amplitude set by the initial conditions [m]
- ϕ: phase set by initial displacement and velocity [rad]
- ω_d: damped natural angular frequency [rad/s]
- $\exp(-\zeta \omega_n t)$: exponential envelope that controls decay

The peaks trace an exponential envelope



$$\text{envelope} = \pm A \exp(-\zeta \omega_n t)$$

Peak spacing gives the period. Peak-height reduction gives damping.

Use peaks with the same sign, or use absolute peak magnitudes consistently.

Logarithmic decrement turns decay into a ratio

$$\delta = (1/N)\ \ln(|x_i| / |x_{i+n}|)$$

$$\zeta = \delta / \sqrt{(4\pi^2 + \delta^2)}$$

- x_i and x_(i+N) are peak amplitudes of the same sign.
- N is the number of complete cycles between the selected peaks.
- Using several cycles reduces sensitivity to cursor placement and noise.
- A line fit to ln|peak| versus time provides a stronger check.

28–30 min

Peak timing gives damped and undamped frequency

$$T_d = (t_{i+n} - t_i)/N$$

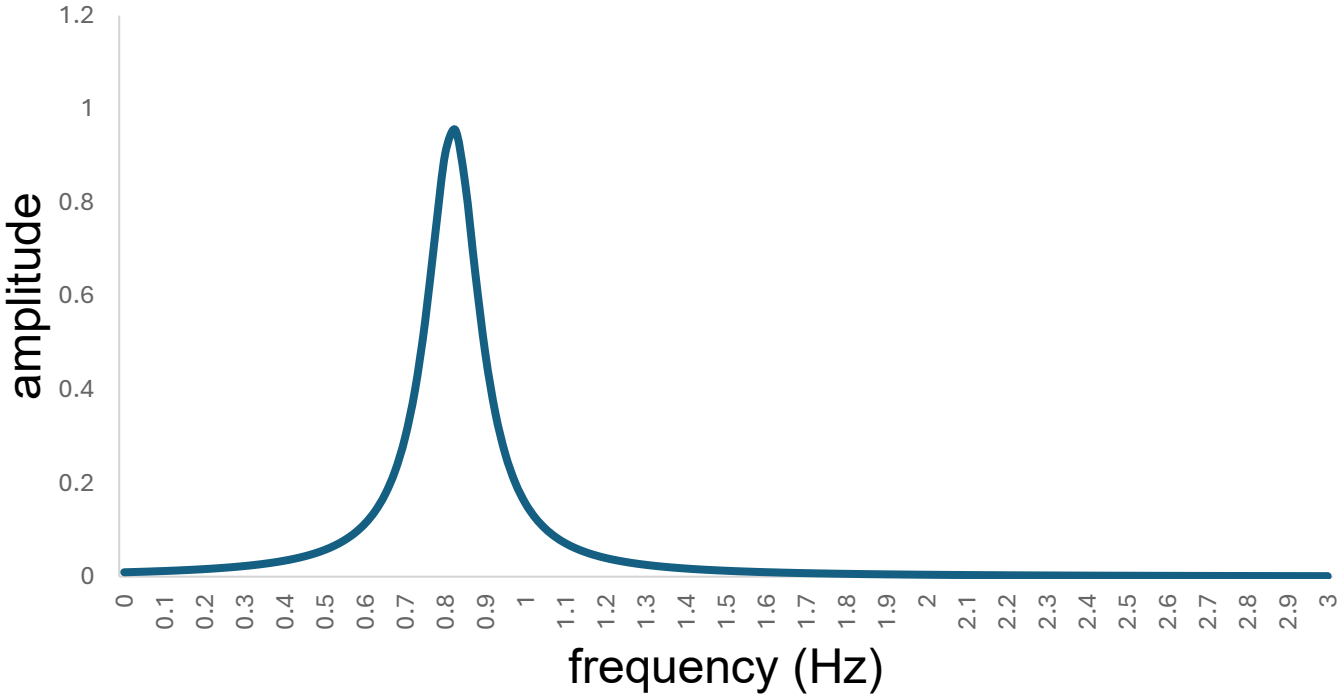
$$f_d = 1/T_d$$

$$\omega_d = 2\pi f_d$$

$$\omega_n = \omega_d / \sqrt{(1 - \zeta^2)}$$

For light damping, f_d and f_n are close, but they are not identical. Report which one you measured and which one you calculated.

The FFT finds the frequency that best matches the record



largest spectral peak

$$f_{peak} \approx fd$$

The cart repeats most strongly at its damped natural frequency, so the correlation with that sinusoid becomes largest.

Time-domain and FFT estimates must agree

peak spacing

$f_{d,time}$

direct and intuitive

FFT peak

f_{peak}

fast visual check

model fit

f_d, ζ

uses the entire record

If the estimates disagree, inspect sampling, drift, peak selection, units, and whether the record contains more than one motion.

Spring stiffness comes from a separate free-vibration test

$$k \approx m_h (2\pi f_s)^2$$

- Suspend one spring with a known moving mass m_h.
- Give a small vertical displacement and record several cycles.
- Use Graphical Analysis sinusoidal fit or exported force-time data to obtain f_s.
- Repeat with the two largest safe masses if time permits.
- State whether the hanger/attachment mass is included in m_h.

Nominal value

10.7 N/m

Measure it.
Do not assume it.

Two springs act together on the cart

For a cart between a left and right spring, a small displacement stretches one spring and compresses the other.

$$k = k_L + k_R$$

$$f_{n,theory} = (1/2\pi)\sqrt{(k/m)}$$

Use the total moving mass, including the cart, added bars, and mounted dampers.

The seven-case matrix isolates one cause at a time

case	moving mass	spring pair	dampers	purpose
1	base	base	0	reference
2	+ mass	base	0	mass effect
3	++ mass	base	0	mass trend
4	base	base	2	damping effect
5	base	base	4	damping trend
6	base	softer pair	0	stiffness effect
7	base	stiffer pair	0	stiffness trend

Collection settings must preserve the motion

sample rate — **≤ 25 Hz**

enough samples per cycle

record length — **until nearly at rest**

captures the decay envelope

initial displacement — **measure and record**

release without a push

position zero — **set before each run**

prevents avoidable offsets

A reliable run follows the same sequence every time

1. Level track and confirm clear travel
2. Pair the cart by serial number
3. Zero position at equilibrium
4. Displace a measured distance
5. **Click Collect, then release without push**
6. Record until motion nearly stops
7. Export CSV with a descriptive file name
8. Plot immediately and repeat if needed

The analysis code produces four pieces of evidence

1

clean x(t)

remove offset and inspect drift

2

peak table

times, amplitudes, and cycle spacing

3

decay estimate

δ, ζ, and fitted envelope

4

frequency check

time-domain f_d and FFT peak

Keep raw CSV files unchanged. The code writes figures and a summary table to a separate results folder.

Quality checks catch most bad runs before students leave

Accept the run when...

- several complete cycles
- smooth exponential-like envelope
- cart remains on track
- no clipped or missing samples
- metadata matches file name

Repeat the run when...

- push at release
- spring goes slack or overstretches
- cart or cable rubs
- track not level
- wrong mass or spring recorded

Fully analyze one case and show the result to the GTA before leaving.

Pre-Lab 4 prepares the decisions made at the bench

- Identify the industrial role of free-response testing.
- Calculate ω_n, ζ, and ω_d for a new spring–mass–damper case.
- Predict how mass, stiffness, and damping change the record.
- Run provided code for an analytical response and a numerical solution.
- Use provided peak-analysis code to estimate damping and frequency.
- Submit one Word or PDF file with answers and required figures before your own Lab 4 section begins.

No Python or MATLAB programming is required. Students run supplied code, interpret outputs, and explain the physics.

Lab 4 evidence feeds the combined Labs 4–5 report

Model

define m, k_eq, c, ω_n, ζ, and assumptions

Method

describe calibration, cases, sampling, and release

Results

time traces, spectra, parameter table, uncertainties

Discussion

compare trends with theory and diagnose deviations

The central claim should answer: do the measured changes in natural frequency and damping follow the spring–mass–damper model?